Module 10 - Assignment & Relational Operators

1. What are Assignment Operators?

Assignment Operators are used to assign or update values in a variable. The basic assignment operator is =.

Example Code:

```
// Assignment Operators
let age = 20;
let additionalVal = 100;

// Adds a value to the variable and assigns the result.
age += 20;             // age = age + 20 ==> 20 + 20 = 40
age += additionalVal; // age = age + additionalVal ==> 40 + 100 = 140

// Subtracts a value from the variable and assigns the result.
age -= 10;             // age = age - 10 ==> 140 - 10 = 130

// Multiplies the variable by a value and assigns the result.
age *= 2;              // age = age * 2 ==> 130 * 2 = 260

// Divides the variable by a value and assigns the result.
age /= 2;              // age = age / 2 ==> 260 / 2 = 130

// Assigns the remainder of the division to the variable.
age %= 2;              // age = age % 2 ==> 130 % 2 = 0

// Raises the variable to the given power and assigns the result.
age **= 2;             // age = age ** 2 ==> 0 ** 2 = 0

console.log(age);
```

2. What are Relational Operators?

Relational Operators, also called Comparison Operators, are used to compare two values and return a Boolean result (true or false).

Example Code:

```
// Relational or Comparison Operators
// Less than
console.log(20 < 20);           // false

// Less than or equal to
console.log(21 <= 20);          // false

// Greater than
console.log(40 > 40);           // false

// Greater than or equal to
console.log(40 >= 39);          // true

// Equal to
console.log(40 == '40');        // true
console.log(40 == 40);          // true

// Not equal to
console.log(40 != '50');        // true
console.log(40 != 50);          // true

// Strict Equal to
// Equal value and same type
console.log(40 === '40');       // false
console.log(40 === 40);         // true

// Strict Not Equal to
// Not equal value and Not same type
console.log(40 !== '40');       // true
console.log(40 !== 40);         // false
```

3. **What is the difference between == and ===?**
   - **== (Loose Equality)** → Compares only value and allows type conversion.
   - **=== (Strict Equality)** → Compares value and data type, without type conversion.

   Example: 40 == "40" → true, but 40 === "40" → false.

4. **What is the difference between != and !==?**
   - **!= (Loose Inequality)** → Compares only value and allows type conversion.
   - **!== (Strict Inequality)** → Compares value and data type without type conversion.

   Example: 40 != "40" → false, but 40 !== "40" → true.